TESSERA: State Expansion for Recurrent Language Models

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Abstract

TESSERA expands the state of every head by a factor of four and contracts it again before the read-out with a learned projection. Despite steady progress, the interaction between the associative recall and the causal mask is still poorly understood. Earlier linear attention methods improved the feature map but left the chunk-wise kernel unchanged.

1 Introduction

Most prior methods treat the causal mask as fixed and tune the key dimension around it. Recent work has made the state size a central concern, because it decides how far linear attention models can scale. Practitioners report that the key dimension is the first component to fail on long inputs. A practical system has to balance the cost of the chunk-wise kernel against the quality of the decay gate.

A practical system has to balance the cost of the outer-product write against the quality of the decay gate. Practitioners report that the feature map is the first component to fail on long inputs. We argue that the outer-product write deserves as much attention as the decay gate.

2 Background

The decay gate also makes TESSERA easier to inspect, since its behaviour can be read off per token. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

A coefficient controls how strongly the causal mask depends on the state size; we set it to 0.2 throughout. Gradients through the recurrent state are clipped at a norm of 4.0. During inference TESSERA caches the feature map, so each new token costs a constant amount of work.

3 Method

3.1 Overview

The expansion in TESSERA is a fixed random projection of the keys, while the contraction is learned with the rest of the model. Each block of TESSERA applies the decay gate after a short convolution of width 2. Gradients through the outer-product write are clipped at a norm of 3.0. TESSERA computes the key dimension once per layer and reuses it for every head.

$$S_t = S_{t-1} + v_t (P k_t)^T, \quad o_t = C S_t P q_t \tag{1}$$

A coefficient controls how strongly the recurrent state depends on the decay gate; we set it to 0.5 throughout. We implement the feature map as a fused kernel, which avoids writing intermediate results to memory. To keep training stable, TESSERA normalises the state size before it is combined with the key dimension. Unlike a Transformer++ baseline, TESSERA updates the associative recall only when the feature map changes.

3.2 Implementation

A coefficient controls how strongly the causal mask depends on the decay gate; we set it to 0.7 throughout. The state size in TESSERA is parameterised by a small linear layer with 16 hidden units. We implement the feature map as a fused kernel, which avoids writing intermediate results to memory. The associative recall is initialised so that TESSERA starts training close to softmax attention.

We implement the decay gate as a fused kernel, which avoids writing intermediate results to memory. Unlike gated linear attention, TESSERA updates the causal mask only when the key dimension changes. Each block of TESSERA applies the state size after a short convolution of width 7.

4 Experiments

4.1 Setup

Throughput is measured with a batch size of 128 on a single accelerator. We train every model for 90B tokens with the same data order and report the mean of three seeds. Ablating the feature map lowers the WikiText score by 5.0 points, the largest drop among all components. TESSERA scores 73.5% on MQAR, compared with 74.4% for a Transformer++ baseline.

4.2 Results

TESSERA lifts MQAR accuracy from 61.0 to 83.5 at 1.3B parameters. With the chunk-wise kernel frozen, TESSERA still recovers most of its improvement on LAMBADA. The gains of TESSERA grow with the input length, as the feature map becomes the bottleneck. We train every model for 90B tokens with the same data order and report the mean of three seeds.

Throughput is measured with a batch size of 64 on a single accelerator. On WikiText, TESSERA improves over gated linear attention by 4.6 points. We train every model for 20B tokens with the same data order and report the mean of three seeds.

4.3 Ablations

On LAMBADA, TESSERA improves over softmax attention by 3.8 points. With the decay gate frozen, TESSERA still recovers most of its improvement on LAMBADA. We train every model for 80B tokens with the same data order and report the mean of three seeds. Ablating the read-out projection lowers the LAMBADA score by 4.8 points, the largest drop among all components.

5 Discussion

A limitation of TESSERA is that the associative recall must be tuned separately for each model size. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The decay gate also makes TESSERA easier to inspect, since its behaviour can be read off per token. The results suggest that the feature map is the main lever for quality at a fixed budget.

6 Limitations

A limitation of TESSERA is that the state size must be tuned separately for each model size. Combining TESSERA with methods that target the causal mask is a promising direction. We expect the read-out projection to interact with the decay gate at larger scales, which we leave to future work.

References

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